

Homework #2

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February 25, 2016

Question #1

- Valid Operation
- Valid Operation
- Key Constraint (Key #4 is already taken)
- Key Constraint, Not Null Constraint (Key cannot be null)
- Valid Operation
- Valid Operation
- Integrity Constraint (Did not update WORKS_ON or DEPENDANT)
- Integrity Constraint (Did not update WORK_ON)
- Valid Operation
- Integrity Constraint (Did not update WORKS_ON)
- Valid Operation

Question #2

Grader note: the rho (ρ) is too small in L^AT_EX to be distinguished from a subscript, so instead P is used in it's place.

- $\Pi_{\text{Fname, Minit, Lname}}(\sigma_{\text{Dno} = '5' \wedge \text{hours} = > 10}(\text{project} \bowtie_{\text{Dname} = 'rojectX'} \text{on}))$
- $\Pi_{\text{Fname} = \text{Dependent_Name}}(\text{mployee} \bowtie_{\text{Dependent}})$
- $\Pi_{\text{Fname, LName}}(\text{mployee} \div \Pi_{\text{Super_SSN}})(\sigma_{\text{Fname} = 'Frankli'})$
- $\Pi_{\text{Lname, Sum_Hours}}(s_{\text{name}} \mathcal{F}_{\text{sum hours}}(\text{ROJECT} \bowtie_{\text{ESSN} = \text{SSN}}))$
- $\text{mployee1} \rightarrow \text{mployee} \bowtie_{\text{ESSN} = \text{SSN}} \text{ork_On}$

- f. $\Pi_{Lname}(\Pi_{Lname, SSN} \bowtie \Pi_{Lname, SSN} \text{ employee } \bowtie \Pi_{ESSN = SSN} \text{ orks_On })$
- g. $\Pi_{Fname, Average Salary}(\Pi_{Dname} \mathcal{F}_{average salary} \Pi_{Dno = Dnumber} \text{ employee } \bowtie \text{ Department })$
- h. $\Pi_{Sex} \mathcal{F}_{average salary}(\sigma_{sex = 'female'} \text{ employee })$
- i. $\Pi_{Department1} \bowtie \Pi_{Number = Dnumber} \bowtie \Pi_{Department2}$
- j. $\Pi_{Lname}(\Pi_{Lname, SSN} \bowtie \Pi_{SSN = MGR_SSN} \text{ Department }) - \Pi_{Lname} \text{ employee }$

Question #3

- $\Pi_{ust\#} \text{ ORDER }$
- $\Pi_{rder} \text{ RDER_ITEM, SHIPMENT }$
- $\Pi_{tem\#} \text{ ORDER, TEM }$
- $\Pi_{arehouse\#} \text{ SHIPMENT }$

Assumptions

- Table headers with the same name are not unique.